

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Pugazharasan K	Reg. No.:	192511341
Section / Batch:	B	Date:	22/07/2026

Code of Conduct

I Pugazharasan K(192511341) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Pugazh.K

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

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QUESTIONS

Q1Binary Search
5 marks**Q2**Stack
5 marks**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Stack
5 marks**Q6**Linked List
5 marks**Q7**Stack
5 marks**Q8**Linked List Delta Structures
5 marks**Q9**Linked List
5 marks**Q10**Queue
5 marks**Q1 Binary Search**

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.**Your Fixed Code**

```
1 int binarySearch(int arr[],int n,int key){  
2     int low=0,high=n-1;  
3     while (low<high){  
4         int mid=(low+high)/2;  
5         if (arr[mid]==key)  
6             return mid;  
7         else if (arr[mid]<key)  
8             low=mid+1;  
9         else  
10            high=mid-1;  
11     }  
12     return -1;  
13 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:/OCFFComp/OCFF2/bin/./lib/gcc/  
/x86_64-w64-  
mingw32/10.3.0/../../../../x86_6  
4-w64-mingw32/bin/ld.exe:  
C:/OCFFComp/OCFF2/bin/./lib/comp
```

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[100];  
4 int top = 0;  
5  
6 int pop() {  
7     if (top == 0) {  
8         printf("Underflow");  
9         return -1;  
10    }  
11  
12    top--;  
13    return stack[top];  
14 }  
15  
16 int main() {  
17     stack[0] = 10;  
18     top = 1;  
19  
20     printf("%d", pop());  
21  
22     return 0;  
}
```

Test Input

stdin input (optional)

Output

10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[S];  
int top;  
void init() {  
    top = 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int stack[S];  
4 int top;  
5  
6 void init() {  
7     top = -1;  
8 }  
9  
10 int main() {  
11     init();  
12     printf("%d", top);  
13     return 0;  
14 }
```

Test Input

stdin input (optional)

Output

-1

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Submitted

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QUESTIONS

Q1Binary Search
5 marks**Q2**Stack
5 marks**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Stack
5 marks**Q6**Linked List
5 marks**Q7**Stack
5 marks**Q8**Linked List Delta Structures
5 marks**Q9**Linked List
5 marks**Q10**Queue
5 marks**Q4 Stack**

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){
    while(stack[top] != ' '){
        postfix[j++] = pop();
    }
    pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.**Your Fixed Code**

```
1 #include <stdio.h>
2
3 char stack[100];
4 char postfix[100];
5 int top = -1;
6 int j = 0;
7
8 char pop() {
9     return stack[top--];
10 }
11
12 void check(char ch) {
13     if (ch == ' ') {
14         while (top >= 0 && stack[top] != ' ') {
15             postfix[j++] = pop();
16         }
17         if (top >= 0) {
18             pop();
19         }
20     }
21 }
22
23
```

Test Input

stdin input (optional)

Output Pending manual review

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 char stack[100];  
4 int top = -1;  
5  
6 char pop() {  
7     return stack[top--];  
8 }  
9  
10 int check(char ch) {  
11     if ((ch == '[' && stack[top] == '[') ||  
12         (ch == ']' && stack[top] == '[')) {  
13         pop();  
14         return 1;  
15     }  
16     return 0;  
17 }  
18  
19 int main() {  
20     stack[++top] = '[';  
21     printf("%d", check('['));  
22  
23 }
```

Test Input

stdin input (optional)

Output

1

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SIMATS Online Programming Lab

DS PROGRAMMING

PUGAZHARASAN K
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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q6 Linked List

5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *head = NULL;
11     struct Node *temp;
12
13     head = (struct Node *)malloc(sizeof(struct Node));
14     head->data = 10;
15     head->next = NULL;
16
17     temp = head;
18
19     while (temp != NULL) {
20         printf("%d ", temp->data);
21         temp = temp->next;
22     }
23
24     return 0;
25 }
```

Test Input

stdin input (optional)

Output

10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int stack[100];
4 int top = 0;
5
6 int POP() {
7     if (top == 0) {
8         printf("Underflow");
9         return -1;
10    } else {
11        top--;
12        return stack[top];
13    }
14 }
15
16 int main() {
17     stack[0] = 10;
18     top = 1;
19     printf("%d", POP());
20
21
22     return 0;
23 }
```

Test Input

stdin input (optional)

Output

10

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QUESTIONS

Q1Binary Search
5 marks**Q2**Stack
5 marks**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Stack
5 marks**Q6**Linked List
5 marks**Q7**Stack
5 marks**Q8**Linked List Data Structures
5 marks**Q9**Linked List
5 marks**Q10**Queue
5 marks**Q8 Linked List Data Structures**

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.**Your Fixed Code**

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 int main() {
10     struct Node *current, *prev, *next;
11
12     current = NULL;
13     prev = NULL;
14
15     while (current != NULL) {
16         next = current->next;
17         current->next = prev;
18         prev = current;
19         current = next;
20     }
21
22     return 0;
23 }
```

Test Input

stdin input (optional)

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QUESTIONS

Q1Binary Search
5 marks**Q2**Stack
5 marks**Q3**Stack
5 marks**Q4**Stack
5 marks**Q5**Stack
5 marks**Q6**Linked List
5 marks**Q7**Stack
5 marks**Q8**Linked List Delta Structures
5 marks**Q9**Linked List
5 marks**Q10**Queue
5 marks**Q9 Linked List**

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new.next = head;  
2 head = new;
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_19251134  
1RB_6a794e6510d937.47455046\main  
.crl:4: error: expected '(',  
'.', '?', 'asm' or  
'__attribute__' before '1 + 1' token
```

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QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Delta Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

DEQUEUE(Q):

if front == -1:

print("Underflow")

else:

return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 int Q[5] = {10, 20, 30, 40, 50};
4 int front = 0;
5
6 int DEQUEUE() {
7     if (front == -1) {
8         printf("Underflow");
9         return -1;
10    } else {
11        int item = Q[front];
12        front++;
13        return item;
14    }
15 }
16
17 int main() {
18     printf("%d", DEQUEUE());
19     return 0;
20 }
```

Submitted

Test Input

stdin input (optional)

Output

10

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (expression[i] != '\0') {  
2     if (isdigit(expression[i]))  
3         push(expression[i] - '0');  
4     else {  
5         int op2 = pop();  
6         int op1 = pop();  
7         int result;  
8         switch (expression[i]) {  
9             case '+': result = op1 + op2; break;  
10            case '-': result = op1 - op2; break;  
11            case '*': result = op1 * op2; break;  
12            case '/': result = op1 / op2; break;  
13        }  
14        push(result);  
15    }  
16    i++;  
17 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_19251134  
1RB_6e794ed8263fc2.06259824\main  
.c:1:1: error: expected  
identifier or '(' before 'while'  
1 while (expression[i] != '\0') {  
  ^
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Stack
5 marks

Q9
Circular Queue
5 marks

Q10
Stack
5 marks

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1)
3         return -1;
4
5     int data = queue[front];
6
7     if (front == rear) {
8         front = -1;
9         rear = -1;
10    } else {
11        front = (front + 1) % MAX;
12    }
13
14    return data;
15 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\wandbox_19251134
1RB_6a794f23aff09_37359421\main
.c: In function 'dequeue':
C:\Windows\TEMP\wandbox_19251134
1RB_6a794f23aff09_37359421\main

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {  
2     if (l > r)  
3         return -1;  
4  
5     int mid = l + (r - l) / 2;  
6  
7     if (arr[mid] == x)  
8         return mid;  
9  
10    if (arr[mid] > x)  
11        return binarySearch(arr, l, mid - 1, x);  
12  
13    return binarySearch(arr, mid + 1, r, x);  
14 }
```

Test Input

stdin input (optional)

Output

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Submitted

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {  
2     if (front == 0)  
3         front = MAX - 1;  
4     else  
5         front = front - 1;  
6  
7     queue[front] = data;  
8 }
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1Stack Notations - Postfix
Expression
5 marks**Q2**Circular Queue
5 marks**Q3**Binary Search
5 marks**Q4**Queue
5 marks**Q5**Queue
5 marks**Q6**Doubly Linked List
5 marks**Q7**Stack
5 marks**Q8**Stack
5 marks**Q9**Circular Queue
5 marks**Q10**Stack
5 marks**Q5 Queue**

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.**Your Fixed Code**

```
1 void enqueue(int data) {  
2     if (rear == MAX - 1) return;  
3     queue[++rear] = data;  
4 }
```

Test Input

stdin input (optional)

Output Pending manual review Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node* del) {  
2     if (del->prev != NULL)  
3         del->prev->next = del->next;  
4     else  
5         head = del->next;  
6  
7     if (del->next != NULL)  
8         del->next->prev = del->prev;  
9 }
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if top == 0 → Underflow
2 else
3     POP()
```

Test Input

stdin Input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Stack

5 marks

Point out the error.

if top == MAX:

Overflow

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top == MAX) {  
2     printf("Overflow");  
3 } else {  
4     push();  
5 }
```

Test Input

stdin input (optional)

Output

Pending manual review

Submitted

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q6

Doubly Linked List
5 marks

Q7

Stack
5 marks

Q8

Stack
5 marks

Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;  
queue[rear] = vehicle;  
if(rear == front){  
    printf("Queue Full");  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((rear + 1) % MAX == front) {  
2     printf("Queue Full");  
3 } else {  
4     rear = (rear + 1) % MAX;  
5     queue[rear] = vehicle;  
6 }
```

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Q9

Circular Queue
5 marks

Q10

Stack
5 marks

Q10 Stack

5 marks

Point out the error?

PEEK(stack):

return stack[top+1]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 PEEK(stack);  
2 return stack[top];
```

Test Input

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SIMATS Online Programming Lab

DS PROGRAMMING

PUGAZHARASAN K
192511341RB

0/5 answered

CO3_AT3_MCQ OPEN

Multiple Choice Questions

0/25 answered

Attempt →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

Debugging & Optimization Task

10/10 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →